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Research paper

Semantic-geometric dual knowledge guided instance segmentation for vehicle components

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ABSTRACT

The detection and segmentation of vehicle components are crucial steps in intelligent vehicle damage assessment. However, due to the wide variety of vehicle components with diverse shapes and the high similarity between mirror-symmetric components, missed detections and false positives remain common challenges in vehicle component detection and segmentation. To address these challenges, this paper proposes a deep learning-based dual-knowledge guided vehicle component instance segmentation network. The proposed method fuses implicit multi-scale semantic knowledge by the enhanced semantic knowledge network, thereby improving the model's capability to filter and capture critical component information. This design enhances focus on key foreground features while suppressing interference from background features. Furthermore, the proposed embedded geometric knowledge module synergizes geometric constraint knowledge of vehicle components with the model's intrinsic learning capability. By reasoning about positions among three types of landmark components, it explicitly supplements the model with spatial relationship information of components that are inherently challenging to learn from data alone. We evaluate the detection and segmentation performance of the proposed method on a dataset comprising 59 classes of vehicle components. Compared with the baseline model, our method achieves significant improvements of 19.3% and 19.1% in detection mean average precision and segmentation mean average precision, respectively. Additionally, we evaluate the generalization capability of the proposed method on an independent public dataset. Compared with state-of-the-art instance segmentation methods, the proposed method demonstrates superior performance in both detection and segmentation tasks for vehicle components.

1. Introduction

The increasing prevalence of private vehicles in modern society has led to an inevitable rise in traffic accident frequency. Vehicle damage assessment (Hsu et al., 2015), a critical process of post-accident insurance claim settlement, currently relies predominantly on manual on-site inspection methods that suffer from high operational costs, inconsistent standards, and low efficiency. In this regard, integrating deep learning (Fan et al., 2024; Nanayakkara and Meegama, 2024) with damage assessment processes enables intelligent damage assessment (Zhang et al., 2020; Maiano et al., 2023), which utilizes instance segmentation to simultaneously detect and segment both damage types and corresponding vehicle components, significantly enhancing assessment efficiency. The intelligent damage assessment workflow proceeds as follows: vehicle images are uploaded to the assessment system, which automatically detects and segments both damage classes and corresponding components, subsequently generating a compensation proposal. In this process, accurate and efficient instance segmentation of vehicle components serves as the fundamental prerequisite for implementing intelligent damage assessment.

Fig. 1 illustrates typical images used in vehicle damage assessment, where each image typically contains dozens of component classes. The multi-object nature presents the first major challenge for vehicle component instance segmentation (Challenge 1). As demonstrated by the comparative analysis between Fig. 1(a) and Fig. 1(b), identical object class exhibit significant variations in both color and shape features

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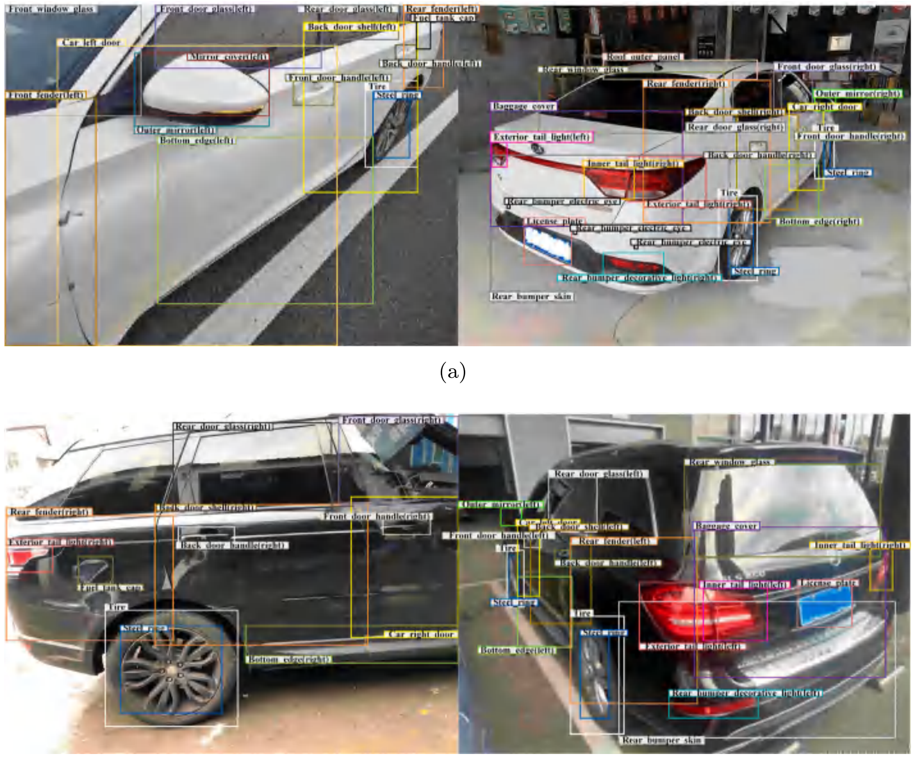


(a)

(b)

Fig. 1. The images used for vehicle damage assessment: (a) shows the left view of the white vehicle and the rear view of the white vehicle; (b) shows the right view of the black vehicle and the rear view of the black vehicle.

across different vehicle images. This phenomenon of low intra-class similarity presents a fundamental challenge for instance segmentation of vehicle components (Challenge 2). Figs. 1(a) and 1(b) also demonstrate that the mirror-symmetry characteristics of vehicles lead to confusion between symmetric classes (e.g., the left door and the right door), highlighting the challenge of high inter-class similarity in vehicle component instance segmentation (Challenge 3).

Earlier, scholars utilized traditional image processing algorithms for vehicle component detection (Leung, 2004; Chávez-Aragón et al., 2011). These algorithms achieved image extraction of vehicle components by learning the contour features of each component, enabling detection in simple backgrounds. However, traditional image processing algorithms commonly suffer from poor applicability, low accuracy, and limited robustness, rendering them unsuitable for detecting multiple vehicle components in complex backgrounds. The introduction of deep learning algorithms garnered widespread attention, leading to the gradual emergence of vehicle component detection and segmentation methods based on deep learning (Lu et al., 2014; Yu, 2017; Shu, 2017).

However, existing vehicle component detection and segmentation methods are primarily designed for scenarios with limited component classes, thus demonstrating suboptimal performance in intelligent damage assessment applications. Furthermore, these methods fail to adequately address the widespread issue of low intra-class similarity and high inter-class similarity in the field of intelligent vehicle damage assessment. To address these challenges, as illustrated in Fig. 2, we propose the Dual-Knowledge Guided vehicle component instance segmentation Network (DKGNet) to specifically overcome these three fundamental challenges. This work makes the following three contributions:

(1) A semantic and geometric dual-knowledge guided vehicle component instance segmentation network named DKGNet is proposed, which achieves accurate detection and segmentation of multi-object vehicle components, laying the foundation for the design and application of intelligent vehicle damage assessment systems.

(2) We propose an Enhanced Semantic Knowledge Network (ESKN), which simultaneously learns semantic information from different channels and spatial positions while achieving multi-scale semantic knowledge fusion. This approach effectively minimizes information loss while guiding the model to focus on critical discriminative features and intra-scale features of objects.

(3) We propose an Embedded Geometric Knowledge Module (EGKM), which formally encodes geometric spatial knowledge of vehicle components into the model architecture. By integrating this domain-specific knowledge with the model's inherent learning capabilities, EGKM effectively guides accurate classification of mirror-symmetric vehicle components.

This article is organized as follows. Section 2 presents an overview of the related works. Section 3 provides the details of the proposed methods. Section 4 presents the experimental results. Finally, Section 5 concludes this article.

2. Related work

2.1. Detection and segmentation of vehicle components based on deep learning

Current deep learning-based vehicle component detection and segmentation methods have achieved good results when applied to a small number of vehicle component classes (Lu et al., 2014; Yu, 2017; Shu, 2017; Xie, 2019; Wu, 2020). Works (Qianqian et al., 2019; Singh et al., 2019) proposed Mask R-CNN-based vehicle component instance segmentation methods to address localization errors caused by component occlusion. Chang (Chang, 2019) utilized symbiotic relationships between vehicle components to construct a component recognition network through semantic feature merging. Work (Barari et al., 2020) performed semantic segmentation of vehicle components and designed simple front-rear markers to correct easily confused left–right component classes. Work (Pasupa et al., 2022) compared the performance

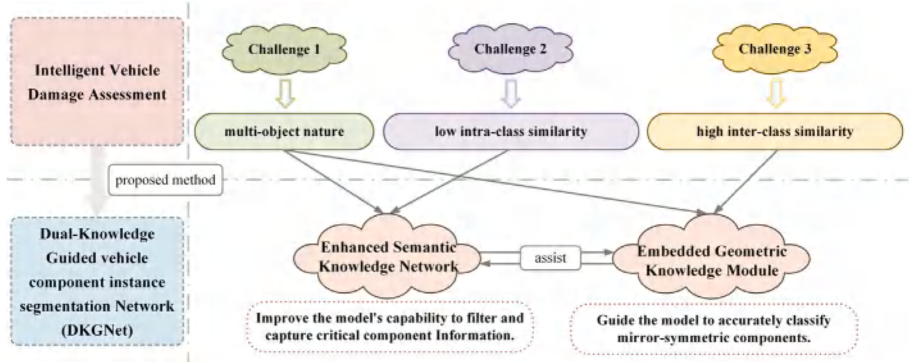


Fig. 2. The existing challenges and solutions.

of different deep learning algorithms in vehicle component instance segmentation, while also testing algorithm robustness under varying weather and lighting conditions. However, previous studies focused on limited vehicle models and component classes, requiring further expansion. To address this, work (Abhilash et al., 2023) adopted a weakly semi-supervised component segmentation technique combined with an online label optimization strategy to expand the dataset. Work (Xu et al., 2023) utilized a GAN generator and inverter for data generation, employing a label generator to produce pixel-level labels through hypercolumn representations in latent space. Recently, for multi-object vehicle component detection and segmentation tasks, SU-VPDN (Zhai et al., 2024a) transformed the challenge of detecting similar objects into directional scene detection by constructing three subnetworks – original vehicle scene classification, vehicle component instance segmentation, and scene understanding – and integrating their results to achieve accurate vehicle component recognition. MTFDN (Zhai et al., 2024b) proposed a decoupled network architecture, designing parallel dynamic attention modules, bidirectional fused feature pyramid networks, and bidirectional hybrid dilated fused feature pyramid networks, then integrating these three improved modules into the architecture to enhance the model's perception of vehicle components in complex backgrounds.

Although the aforementioned studies share similarities with our work, they focus on limited component classes and single image types, making them inapplicable for intelligent damage assessment. In contrast, our work focuses on both the intra-class variations and inter-class similarities among numerous vehicle component classes. By fully leveraging the semantic knowledge from images and the geometric knowledge of components, we enable mutual reinforcement between these two aspects, successfully achieving accurate detection and segmentation of multi-object vehicle components.

2.2. Knowledge-guided detection and segmentation

Since purely data-driven methods cannot achieve satisfactory results, an increasing number of studies (Guo et al., 2018; Pouyanfar et al., 2018; Chawla et al., 2002; Wu et al., 2025; Wang et al., 2025) have incorporated domain knowledge guidance alongside data-driven approaches. Domain knowledge refers to accumulated experience, patterns, and professional expertise in a specific field, which can be obtained through mining and extracting data features or by summarizing empirical experience. Existing research has demonstrated that combining domain knowledge with deep learning models can effectively enhance a model's understanding of objects. For example, Jiang et al. (2018) innovatively combined implicit and explicit knowledge through knowledge routing to enhance the model's capability in handling long-tailed data and semantic consistency. Chen et al. (2018) utilized high-level prediction results as prior knowledge to guide more fine-grained feature learning at lower levels. Work (Ding et al., 2018) proposed a prior knowledge-based deep learning method that improves robots' object recognition ability in complex indoor environments by fusing color and scene prior knowledge. Work (Yang et al., 2018) introduced a semantic navigation method based on graph convolutional networks and deep reinforcement learning, which constructs a semantic knowledge graph to encode functional spatial relationships between objects. Work (Hu et al., 2018) proposed an object relation module that models inter-object relationships through interactions of appearance features and geometric positions. Liu et al. (2018) transformed the detection task into a graph structure reasoning problem by modeling scene context and inter-object relationships, thereby reducing illogical detection results. Xu et al. (2019) developed a spatially-aware graph relation network to adaptively discover and integrate critical semantic and spatial relationships for reasoning about each object. Work (Chen et al., 2020) constructed a structured knowledge graph based on statistical label co-occurrence data to explicitly model label correlations, enhancing the modeling of label dependencies and few-shot generalization capability.

The aforementioned studies demonstrate that domain knowledge can guide deep learning algorithms to achieve more accurate object recognition in complex scenarios, thereby reducing both missed detections and false positives. Therefore, to address the research challenges of low intra-class similarity and high inter-class similarity among vehicle components, we investigate the characteristics of vehicle component semantic knowledge and its fusion methods, while also exploring the acquisition and application of geometric knowledge about vehicle components. This enables knowledge-guided instance segmentation of vehicle components.

3. Method

To address the current challenges in vehicle component instance segmentation, this paper proposes a semantic and geometric dual-knowledge guided vehicle component instance segmentation network — DKGNet. Its architectural framework is shown in Fig. 3. Specifically, from the semantic perspective, to address the challenges of low intra-class similarity and high inter-class similarity among vehicle components, DKGNet introduces the Enhanced Semantic Knowledge Network (ESKN) following the Convolutional Neural Network (CNN) (Krizhevsky et al., 2012). The ESKN module enhances the model's ability to learn key features and intra-scale features while minimizing information loss through multi-scale semantic fusion. From the geometric perspective, to specifically tackle the high similarity between mirror-symmetric vehicle components, DKGNet incorporates the Embedded Geometric Knowledge Module (EGKM) in the classification and regression pipeline. The EGKM formalizes and embeds the inherent geometric spatial knowledge of vehicle components into the model, thereby utilizing domain-specific knowledge to significantly improve the model's discrimination capability for mirror-symmetric components.

The ESKN module provides stronger semantic information support for subsequent tasks through multi-scale semantic fusion. The EGKM

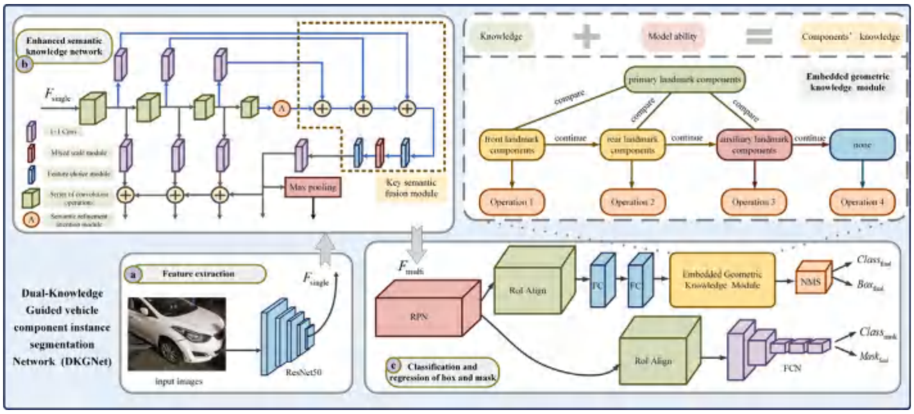


Fig. 3. Overall framework of the proposed DKGNet.

module further embeds geometric prior knowledge of vehicle components based on the enhanced features from ESKN, thereby improving the model's ability to distinguish left and right components. These two modules complement each other from the perspectives of semantic enhancement and geometric reasoning, jointly enhancing the model's performance.

Algorithm 1: Dual-knowledge guided vehicle component instance segmentation.

Input:

I_{vehicle}: the image fed into the network ;

Output:

Box_{final}: the final predicted bounding boxes ;

$Class_{\text{final}}$: the class scores corresponding to the predicted bounding boxes ;

$Mask_{\text{final}}$: the final predicted masks ;

$F_{\text{single}} \leftarrow \text{CNN}_{\text{ResNet50}}(I_{\text{vehicle}})$;

$F_{\text{multi}} \leftarrow \text{ESKN}(F_{\text{single}})$: Through multi-scale semantic knowledge fusion, key information is filtered and captured;

$[P, Class_{\text{RPN}}] \leftarrow \text{RPN}(F_{\text{multi}})$;

if *detection* **then**

  $RoI_{7\times7} \leftarrow \text{RoIAlign}_{7\times7}(P, F_{\text{multi}})$;

  $[Box_{\text{initial}}, Class_{\text{initial}}] \leftarrow \text{FC}(RoI_{7\times7})$: The preliminary predicted bounding boxes and their classes are obtained through the classifier;

  $[Box_{\text{final}}', Class_{\text{final}}'] \leftarrow \text{EGKM}(Box_{\text{initial}}, Class_{\text{initial}})$: The embedded geometric knowledge guides the model to accurately identify left–right symmetrical components;

  $[Box_{\text{final}}, Class_{\text{final}}] \leftarrow \text{NMS}(Box_{\text{final}}', Class_{\text{final}}')$: Eliminate redundant bounding boxes;

end

else if *segmentation* **then**

  $RoI_{14\times14} \leftarrow \text{RoIAlign}_{14\times14}(P, F_{\text{multi}})$;

  $[Mask_{\text{final}}, Class_{\text{mask}}] \leftarrow \text{FCN}(RoI_{14\times14})$;

end

return Box_{final}, $Class_{\text{final}}$, $Mask_{\text{final}}$

The overall algorithm flow of DKGNet is presented in Algorithm 1. Firstly, the vehicle image is fed into the CNN to extract a single-scale feature map F_{single}. Subsequently, F_{single} is passed through the ESKN to obtain multi-scale feature F_{multi} containing key information. F_{multi} serves as the input to the classification and regression network for object bounding boxes and masks. Initially, F_{multi} is processed by the Region Proposal Network (RPN) (Ren et al., 2015) module to extract region proposals P and the corresponding class scores $Class_{\text{RPN}}$. Afterwards, F_{multi} and P are jointly passed into two Region of Interest Align (RoI Align) (He et al., 2017) structures, generating feature $RoI_{7\times7} \in \mathrm{R}^{7\times7}$ and feature $RoI_{14\times14} \in \mathrm{R}^{14\times14}$, respectively. In the detection branch, $RoI_{7\times7}$ is processed by Fully Connected (FC) layers to obtain preliminary predicted bounding boxes and their classes. Following that, EGKM is used to make more precise adjustments to the preliminary predicted classes. Non-Maximum Suppression (NMS) (Neubeck and Van Gool, 2006) is then applied to eliminate redundant bounding boxes after adjustment, resulting in the final predicted bounding boxes Box_{final} and their corresponding class scores $Class_{\text{final}}$. In the segmentation branch, $RoI_{14\times14}$ is processed through a Fully Convolutional Network (FCN) (Long et al., 2015) structure to obtain the predicted masks $Mask_{\text{final}}$ and their corresponding class scores $Class_{\text{mask}}$.

3.1. Enhanced semantic knowledge network

The existing methods struggle to fully capture the key information of vehicle components due to two main reasons. First, the downsampling operations used during model training lead to information loss. Second, during feature extraction, these methods fail to adequately learn the importance of all feature information, causing the model to potentially neglect learning critical feature while instead focusing on confusing or less relevant features.

We aim to enhance the model's perception of vehicle components by reducing the loss of detailed information while enabling the model to accurately identify and capture the key features of various vehicle components. To achieve this, we propose the Semantic Refinement Attention Module (SRAM) and the Key Semantic Fusion Module (KSFM), which facilitate semantic knowledge fusion. By integrating the basic feature pyramid network with the two aforementioned modules, we construct the ESKN, whose architecture is illustrated in Fig. 4. Specifically, the SRAM, designed for the characteristics of vehicle components, enhances the model's ability to learn semantic information across different channels and spatial positions. Meanwhile, the KSFM effectively fuses multi-scale semantic knowledge, thereby reducing information loss while guiding the model to learn both critical object features and intra-scale features. Together, these two modules form ESKN, which significantly improves the accuracy of vehicle component detection and segmentation. Detailed descriptions of SRAM and KSFM are provided in Sections 3.1.1 and 3.1.2, respectively.

3.1.1. Semantic refinement attention module

In the study of instance segmentation for multi-object vehicle components, modeling the inter-component relationships and discriminative features between foreground and background is crucial for model

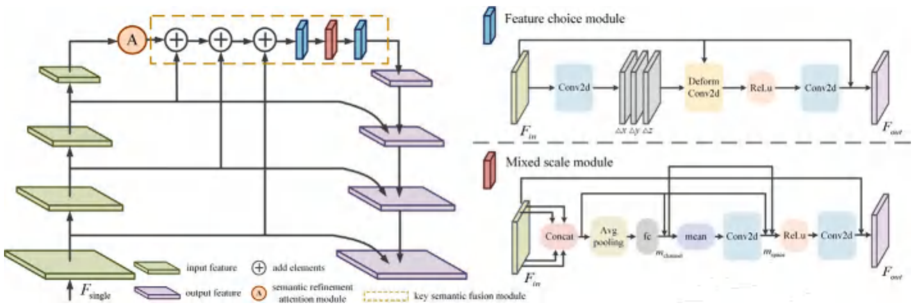


Fig. 4. The structure of enhanced semantic knowledge network.

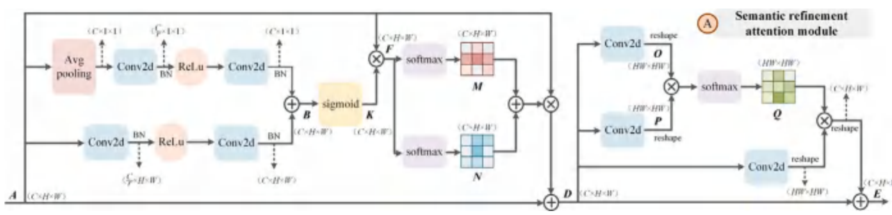


Fig. 5. The structure of semantic refinement attention module.

learning. By incorporating the SRAM, the model can autonomously learn the importance weights of different feature channels, thereby enhancing the representation of relevant vehicle component features while suppressing background interference. Furthermore, the module enables the learning of both horizontal-vertical positional information and spatial relationships among components, ultimately enriching the model's holistic understanding of vehicle components.

As illustrated in Fig. 5, within the SRAM, a given feature $A \in \mathrm{R}^{C\times H\times W}$ is divided into two paths. One path obtains a scalar for each channel through global average pooling, followed by a series of convolution operations to obtain the corresponding feature information. The other path directly undergoes a series of convolution operations to obtain another set of feature information. The convolutional outputs from both pathways are element-wise summed to obtain feature $B \in \mathrm{R}^{C\times H\times W}$. Feature B is processed through a sigmoid operation to obtain the channel-wise attention weights $K \in \mathrm{R}^{C\times H\times W}$. The above process is implemented as follows:

$$B = \mathrm{Co}(\mathrm{Re}(\mathrm{Co}(\mathrm{Avg}(A)))) + \mathrm{Co}(\mathrm{Re}(\mathrm{Co}(A))) \tag{1}$$

$$K = \frac{1}{1 + \exp(-B)} \tag{2}$$

Where Co represents the convolution operation. Re denotes the Rectified Linear Unit (ReLU) operation. Avg represents the average pooling operation.

Then, feature A is directly multiplied by weight K to obtain the initially modified feature $F \in \mathrm{R}^{C\times H\times W}$. A softmax operation is performed on feature F in the horizontal dimension and vertical dimension to obtain the horizontal position weight $M \in \mathrm{R}^{C\times H\times W}$ and vertical position weight $N \in \mathrm{R}^{C\times H\times W}$, respectively, to enable the model to better learn the relationship between the horizontal and vertical features in the image:

$$M = \sigma(A \times K, \mathrm{dim} = -1) \tag{3}$$

$$N = \sigma(A \times K, \mathrm{dim} = -2) \tag{4}$$

Where σ indicates the softmax operation, $\mathrm{dim} = -1$ indicates the softmax operation in the last dimension (horizontal dimension), and $\mathrm{dim} = -2$ indicates the softmax operation in the second-to-last dimension (vertical dimension). The larger the value of the corresponding dimension in M and N is, the more similar the horizontal and vertical feature representations of the two positions are, and the greater the correlation between the two positions is.

Then, weight M and weight N are used to modify feature A, and feature $D \in \mathrm{R}^{C\times H\times W}$ is obtained. The implementation process is as follows:

$$D = ((1-\alpha)M + \alpha N) \times A + A \tag{5}$$

Here, α is a control parameter initialized to 0, with its magnitude automatically adjusted during model training. By setting the value of α, the network can focus on the content from initially emphasizing horizontal position features to gradually favoring vertical position features during training. This setting is motivated by the fact that more shallow-level information in our vehicle images is contained in the horizontal position, and the model should initially prioritize learning these shallow-level features. However, as the training progresses, the model's learning capability strengthens, and it should place more emphasis on learning the vertical position features, which contain deeper-level information.

For the model to further learn the relationship between spatial positions in the image, feature D is sent as input into two convolution layers for dimensional transformation, and $O \in \mathrm{R}^{N\times N}$ and $P \in \mathrm{R}^{N\times N}$ ($N = H \times W$) are obtained. After matrix multiplication of O and P is fed into the softmax operation, the spatial position weight $Q \in \mathrm{R}^{N\times N}$ of the features can be obtained:

$$Q = \sigma(O \cdot P) \tag{6}$$

Where σ indicates the softmax operation. The greater the value of Q is, the more similar the spatial features of two positions are represented, and the greater the correlation between them.

Additionally, feature D is input into another convolution layer for dimensional transformation, matrix multiplication is performed on the transformation result and weight Q, and the result is summed with feature D to modify the feature to the final output feature $E \in \mathrm{R}^{C\times H\times W}$:

$$E = \beta Q \cdot D + D \tag{7}$$

Here, β serves as a transformation coefficient initialized to 0, with its magnitude automatically adjusted as training progresses. By setting β, the model can gradually focus on the spatial position information of the image with the deepening of training.

3.1.2. Key semantic fusion module

The detailed network architecture of KSFM is illustrated within the dashed box in Fig. 4. This module enhances semantic knowledge fusion beyond the feature pyramid network, thereby mitigating feature loss and better preserving deep-level features. In standard feature pyramid network, multi-scale features are typically fused through top-down upsampling operations. To address the critical information loss during this fusion process, KSFM introduces a transformation function that aggregates richer low-level semantic information into high-level semantics prior to upsampling, enabling effective multi-scale semantic knowledge fusion.

In addition, to enhance the network's capability in learning intra-scale information from deep features, KSFM processes high-level semantics through stacked Feature Choice Module (FCM) and Mixed Scale Module (MSM). The FCM learns importance weights for spatial features and performs weighted feature allocation, while the MSM captures multi-scale contextual information without altering feature resolution. By alternately incorporating FCM and MSM, KSFM enables mutual reinforcement between these modules, thereby achieving comprehensive extraction of critical intra-scale features. The architectures of FCM and MSM are shown in Fig. 4.

For input feature F_{in} and output feature F_{out}, the implementation process of FCM is as follows:

$$F_{out} = \mathrm{Co}(\mathrm{Re}(\mathrm{DCo}(\mathrm{Co}(F_{in}), F_{in}))) + F_{in} \tag{8}$$

Where Co represents the convolution operation, Re represents the ReLU operation, and DCo represents the deformable convolution operation. The FCM first computes the offset values and offset weights for each spatial position of the input features through convolutional operations. Subsequently, it aggregates the original positions with the computed offsets, multiplies them by the corresponding weights, and obtains the refined feature values. These refined features are then processed through a ReLU activation function and convolutional layers, before being added to the original input features to produce the final output features of FCM.

For input feature F_{in} and output feature F_{out}, the implementation process of MSM is as follows:

$$m_{\text{channel}} = 2 \cdot \mathrm{FC}(\mathrm{Avg}(\mathrm{Concat}(F_{in}))) \tag{9}$$

$$m_{\text{space}} = 2 \cdot \mathrm{Co}(\mathrm{mean}(m_{\text{channel}} \cdot \mathrm{Concat}(F_{in}))) \tag{10}$$

$$F_{out} = \mathrm{Co}(\mathrm{Re}(m_{\text{channel}} \cdot m_{\text{space}} \cdot \mathrm{Concat}(F_{in}))) + F_{in} \tag{11}$$

Where FC represents the full connection operation, Avg denotes the average pooling operation, Concat represents the channel compression operation, and mean is the average operation along the channel. The MSM first compresses each feature channel, unifies the number of channels into the number of output channels, calculates the channel coefficient m_{channel} and spatial coefficient m_{space} through the operations in Eqs. (9) and (10), and then assigns the corresponding importance to different feature channels. Meanwhile, both coefficients are multiplied with the original features, and the resulting features are processed through ReLU activation and convolutional layers before being added to the original input features, ultimately yielding the output features of MSM.

3.2. Embedded geometric knowledge module

In the task of vehicle component instance segmentation, the mirror-symmetric nature of vehicles results in highly similar representations between left–right symmetric components, posing significant challenges to the model's discrimination capability. Although Section 3.1 has enhanced the network's ability to filter and capture key features of vehicle components through semantic knowledge fusion, relying solely on semantic information extraction from images still makes it difficult for the model to accurately distinguish mirror-symmetric left–right components. To address this, we propose the Embedded Geometric Knowledge Module (EGKM) that significantly improves the model's ability to classify symmetric components by integrating spatial geometric knowledge of vehicle components with the network's inherent feature extraction capabilities.

We summarize the geometric knowledge of vehicle components based on human experiential accumulation in component discrimination, and integrate this geometric knowledge with the model's inherent component recognition capability. Specifically, we select the license plate, tire, outer mirror, and fender as primary landmark components; the grille and baggage cover as front-rear landmark components; and the tail light, door, and door handle as auxiliary landmark components. Here, the primary landmark components correspond to those first noticed during human discrimination and serve to determine the vehicle's global orientation; the front-rear landmark components are key elements that play a decisive role in discriminating front-rear orientation; while the auxiliary landmark components are critical for assisting in left–right orientation discrimination.

The selection of the license plate, tire, outer mirror, and fender as primary landmarks is based on their representativeness in human cognition, as these components are more likely to be prioritized in human perceptual processes. More importantly, these four component classes have abundant samples in the dataset, enabling the model to adequately learn their characteristic features. By choosing these representative and model-familiar components as primary landmarks, we can better guide the model to mimic human cognitive patterns. Similarly, we identify components with sufficient samples and high representativeness for front-rear and left–right orientation from the dataset, based on which we determine the composition of front-rear landmarks and auxiliary landmarks.

The selection of these three landmark components is principally motivated by two factors: (1) the network's intrinsic classification capacity constrains the supplementary role of geometric knowledge, and (2) the network exhibits inferior flexibility in knowledge integration and abstraction compared to human cognition. Consequently, the knowledge embedding process necessitates providing explicit discriminative markers to the network.

The EGKM extracts appropriate geometric knowledge from human classification experience of vehicle components and establishes corresponding priority hierarchies for both primary and auxiliary landmark components. Specifically, the priority order for primary landmarks is: license plate >tire >outer mirror >fender; while for auxiliary landmarks it is: tail light >door >door handle. Building upon these priority hierarchies, the EGKM determines vehicle orientation (front/rear/left/right) in images by analyzing spatial relationships between primary landmarks and either front-rear or auxiliary landmarks. This geometric reasoning enables accurate classification of mirror-symmetric vehicle components through spatial knowledge integration.

In the design of the EGKM module, the model first determines whether the vehicle is front-oriented or rear-oriented. If not, it then checks for left or right orientation. If the sequence is reversed, each left–right discrimination would require additional consideration of spatial relationships between front-rear landmarks and primary landmarks, thereby complicating the process. Therefore, among the primary landmark components, the license plate, used for front-rear orientation identification, is assigned the highest priority. The remaining three primary landmark components for left–right orientation discrimination are given lower priorities. Among these three components, the tire exhibits the highest visual similarity across different vehicles, making it the easiest to recognize and thus the highest priority. The outer mirror follows, while the fender is the most challenging to identify and has the lowest priority. Prioritizing more easily recognizable landmark components helps reduce the model's detection and segmentation time, thereby improving efficiency. Similarly, the priorities of auxiliary landmark components are determined by their recognition difficulty.

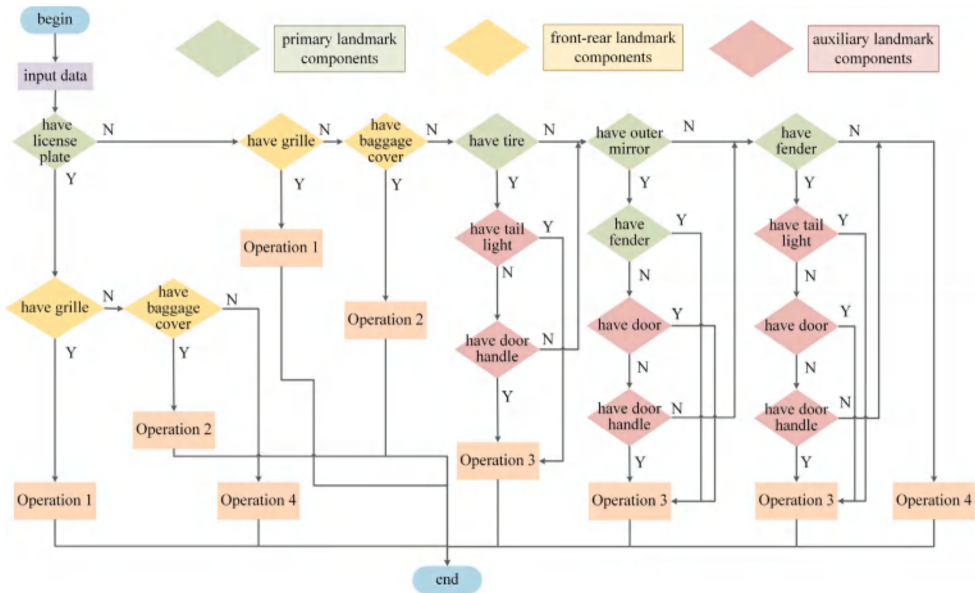


Fig. 6. The flow chart of embedded geometric knowledge module.

Among the three auxiliary landmark components, the tail light has the highest priority due to its frequent appearance in vehicle images and better initial recognition performance. The door follows, while the door handle, being smaller and more challenging for the model to recognize, has the lowest priority.

In the EGKM, the geometric knowledge-guided operations are divided into four parts, corresponding to Operation 1 through Operation 4 in Fig. 6.

Operation 1 and Operation 2 are triggered when the model identifies front-oriented and rear-oriented vehicles, respectively. Operation 1 designates either the license plate or grille as the central component, while Operation 2 uses either the license plate or baggage cover as the central component. Both operations perform left–right component classification by comparing coordinate positions between peripheral components and their central component. For example, when a license plate and a grille are detected in the image, the vehicle's front orientation can be confirmed. Subsequently, the central component (with a horizontal coordinate of x_{cen}) is selected, and Operation 1 is performed as follows:

$$\begin{cases} Labels_{\mathrm{k}} = Labels_{\mathrm{right}}, x_{Labels_{\mathrm{initial}}} < x_{cen} \\ Labels_{\mathrm{k}} = Labels_{\mathrm{left}}, x_{Labels_{\mathrm{initial}}} > x_{cen} \end{cases} \tag{12}$$

Where $Labels_{\mathrm{initial}}$ represents the initial class labels of the left–right symmetric components, $x_{Labels_{\mathrm{initial}}}$ denotes the horizontal coordinates of the left–right symmetric components, $Labels_{\mathrm{right}}$ indicates components labeled as right class, $Labels_{\mathrm{left}}$ signifies components labeled as left class, and $Labels_{\mathrm{k}}$ represents the class labels modified with knowledge.

Operation 3 is executed when the model identifies a left-oriented or right-oriented vehicle. This operation directly guides the network to classify left–right symmetric vehicle components based on spatial relationships between primary and auxiliary landmark components. For instance, when both an outer mirror (with a horizontal coordinate $x_{Lm_{\mathrm{primary}}}$) and a door handle (with a horizontal coordinate $x_{Lm_{\mathrm{auxiliary}}}$) are detected in the image, Operation 3 is performed as follows:

$$\begin{cases} Labels_{\mathrm{k}} = Labels_{\mathrm{right}}, x_{Lm_{\mathrm{auxiliary}}} < x_{Lm_{\mathrm{primary}}} \\ Labels_{\mathrm{k}} = Labels_{\mathrm{left}}, x_{Lm_{\mathrm{auxiliary}}} > x_{Lm_{\mathrm{primary}}} \end{cases} \tag{13}$$

Operation 4 serves as a voting module. When the vehicle image lacks detectable relationships between primary landmarks and either front-rear or auxiliary landmarks, this operation infers left–right classifications for remaining components based on the highest-confidence component's orientation. The complete algorithmic workflow of EGKM is presented in Algorithm 2.

4. Experiments

In this section, the relevant contents of the experiment are introduced, such as the composition of the datasets, the evaluation index, the details of the experiment, the ablation results, the visualization results and the comparison results.

4.1. Datasets

4.1.1. Vehicle multi-part dataset

This study utilizes the Vehicle Multi-Part (VMP) dataset to evaluate the performance of our method. The images in this dataset are captured from the daily insurance business operations of insurance companies, as shown in Fig. 7. It should be noted that all images underwent rigorous anonymization after collection, removing all personally identifiable information (e.g., license plates, faces) and insurer-related sensitive information. The dataset includes images of vehicles of various colors, brands, and types, captured from different views, such as front, rear, left, and right. Most importantly, the dataset contains a significant number of vehicle images with varying degrees of damage, such as deformation, scratches, tear, and breakage, making the VMP dataset perfectly aligned with the application scenarios of vehicle damage assessment.

During the data annotation phase, we collaborated with professional damage assessors from the insurance company to establish annotation guidelines. Following these guidelines, we annotated vehicle components using the Labelme annotation tool, which employs connected points to accurately segment the object and generate the mask information of the vehicle components. The annotations were saved in COCO format as JSON files. Finally, the annotated data underwent sample-based verification by professional assessors to ensure the labeling quality of the VMP dataset.

The VMP dataset contains 45,503 training set images and 11,376 validation set images. According to the requirements of intelligent damage assessment of vehicles, the vehicle component classes in the dataset are divided into 59 classes. Fig. 8 displays the names of 59 component classes, their corresponding label assignments, and the image count per class in the training set. The validation set maintains a 1:4 image ratio

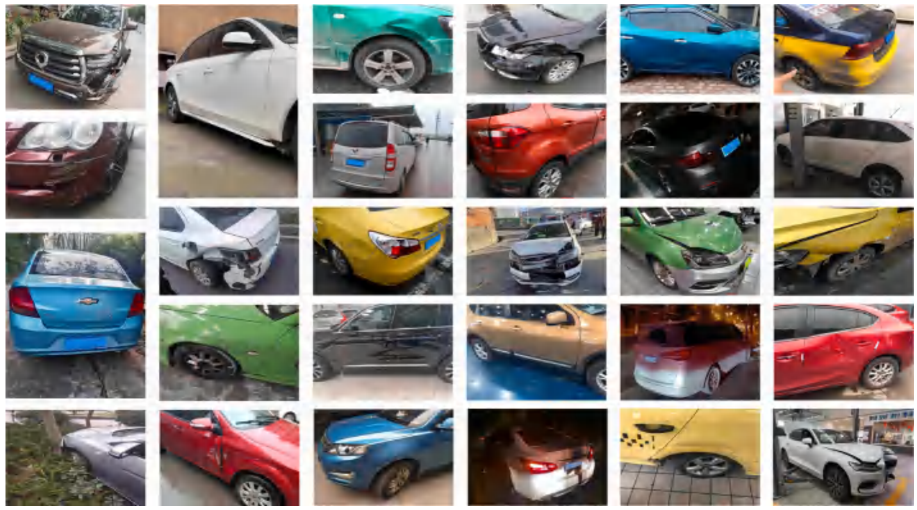


Fig. 7. Representative images from the VMP dataset.

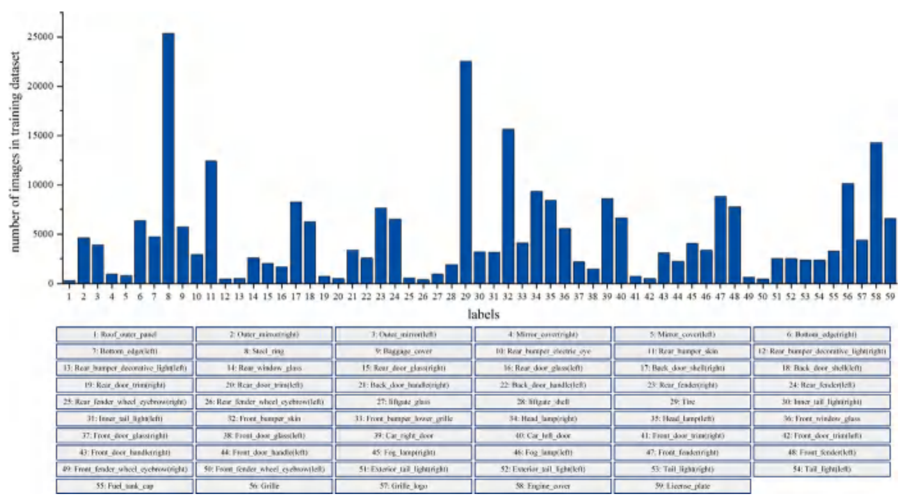


Fig. 8. Component class names, corresponding labels, and their image counts in the training set of the VMP dataset.

to the training set for each component class. Fig. 8 reveals a significant imbalance in sample quantities across different classes within the VMP dataset, exhibiting a long-tail distribution phenomenon. This occurs because the VMP dataset images are entirely collected from the daily insurance operations of insurance companies. In real-world vehicle usage, inherent differences exist in both the damage probability and occurrence frequency of different components. This long-tail distribution precisely reflects the authentic scenario of practical vehicle damage assessment.

4.1.2. DSMLR car part dataset

To further validate the feasibility and generalizability of our method for vehicle component instance segmentation tasks, we additionally employ the public DSMLR car part dataset (Pasupa et al., 2022) (https://github.com/dsmlr/Car-Parts-Segmentation) to evaluate our method.

The dataset encompasses various vehicle types including sedans, pickup trucks, and SUVs, with camera views covering front, rear, and side views. The images are annotated with 18 classes of vehicle components, with left–right symmetric component differentiation. The 18 component classes in the dataset are: back_bumper, back_glass, back_left_door, back_left_light, back_right_door, back_right_light, front_bumper, front_glass, front_left_door, front_left_light, front_right_door, front_right_light, hood, left_mirror, right_mirror, tailgate, trunk, and wheel.

4.2. Evaluation index

To evaluate the superiority of our model and its feasibility for application in an intelligent damage assessment system, we employ two common metrics: Average Precision (AP) and mean Average Precision (mAP), to evaluate the model. These metrics are defined as follows:

$$P_n = \frac{TP(n)}{TP(n) + FP(n)} \times 100\% \tag{14}$$

$$R_n = \frac{TP(n)}{TP(n) + FN(n)} \times 100\% \tag{15}$$

$$AP(n) = \int_0^1 P_n(R_n)\mathrm{d}R_n \tag{16}$$

$$mAP = \frac{1}{S}\sum_{n=1}^{S} AP(n) \tag{17}$$

Where P_n is the precision rate of class n, and R_n is the recall rate of class n. $TP(n)$ indicates that class n is correctly predicted as class n. $FP(n)$ is when the model predicts another class as class n. $FN(n)$ is a case where the model predicts class n to be another class. $AP(n)$ is the average precision of class n, obtained by averaging the P_n of class n under different R_n, which can be used to measure the accuracy of the model's detection and segmentation of various objects. Where S refers to the total number of classes. mAP is the mean average precision,

Algorithm 2: Embedded geometric knowledge module.

Input:

$Labels_{\mathrm{initial}}$: class labels obtained by the network ;
Lm_{primary}: primary landmark components ;
Lm_{front}: front landmark components ;
Lm_{rear}: rear landmark components ;
$Lm_{\mathrm{auxiliary}}$: auxiliary landmark components ;

Output:

$Labels_{\mathrm{k}}$: class labels modified with knowledge ;

```
if Labels_initial in Labels_Lm_primary then
    if Labels_initial in Labels_Lm_front then
        x_cen ← min{x_Lm_primary, x_Lm_front}: Identify the component with the
            smallest horizontal value in the image as the central
            component;
        Labels_k ← Operation1(Labels_initial, x_Labels_initial, x_cen): The labels are
            updated by comparing the horizontal relationship between
            all the components in the image and the central component;
    end
    else if Labels_initial in Labels_Lm_rear then
        x_cen ← min{x_Lm_primary, x_Lm_rear}: Identify the component with the
            smallest horizontal value in the image as the central
            component;
        Labels_k ← Operation2(Labels_initial, x_Labels_initial, x_cen): The labels are
            updated by comparing the horizontal relationship between
            all the components in the image and the central component;
    end
    else if Labels_initial in Labels_Lm_auxiliary then
        Compare the coordinate relationship between the Lm_auxiliary,
            judge the orientation of the vehicle in the image;
        Labels_k ← Operation3(Labels_initial): Modify the initial class
            labels according to the judged orientation;
    end
end
else
    Labels_k ← Operation4(Labels_initial): Modify the initial class labels
        according to the class with the highest confidence in the image;
end
return Labels_k
```

which can be used to measure the overall performance of the model by averaging the AP values for all classes.

4.3. Implementation details

The model described in this paper is trained and tested using an NVIDIA 1080Ti GPU. The operating system is Ubuntu 16.04.6 LTS, and CUDA 11.1 is used for training acceleration. Python 3.8 is used as the programming language, and the PyTorch network development framework is employed.

We use a multi-GPU training method with a batch size of 4 and a stochastic gradient descent algorithm with a momentum of 0.9. The learning rate is dynamically adjusted, starting at 0.008 and reducing to 0.0008 at the 16th epoch, and further dropping to 0.00008 at the 22nd epoch. The Intersection over Union (IoU) threshold of the last NMS module is set to 0.5. The initial values of the control parameter α and the transformation coefficient β in SRAM are both set to 0. All models in this paper utilize random rotatio as a data augmentation method during training, with a rotation probability of 0.5 and an angle range of −15° to 15°. For both training and inference, the shorter edge of input images is fixed at 800 pixels while the longer edge is proportionally scaled (maximum length ≤ 1333 pixels). Zero-padding is applied to handle size inconsistencies within the same batch.

Table 1
Selection of horizontal and vertical weighting coefficients.

α	1-α	mAP_{50}^{box}	mAP_{50}^{mask}	$mAP_{50:95}^{\mathrm{box}}$	$mAP_{50:95}^{\mathrm{mask}}$
0	1	62.4	62.0	51.5	50.1
0.2	0.8	63.3	63.1	52.4	50.9
0.4	0.6	63.8	63.4	52.6	51.1
0.6	0.4	61.0	60.7	50.7	49.5
0.8	0.2	62.4	62.0	51.7	50.1
1.0	0	63.5	63.2	52.2	50.9
Automatic adjustment		**64.8**	**64.4**	**51.7**	**51.1**

Table 2
Ablation study of DKGNet on the VMP dataset.

Method	ESKN	EGKM	mAP_{50}^{box}	mAP_{50}^{mask}	$mAP_{50:95}^{\mathrm{box}}$	$mAP_{50:95}^{\mathrm{mask}}$
Mask R-CNN			53.2	52.7	43.0	42.2
	✓		68.4	67.9	56.6	54.8
		✓	67.4	66.8	54.0	52.6
	✓	✓	**72.5**	**71.8**	**59.9**	**57.6**

4.4. Parameter selection experiments

The parameters α and β in SRAM are updated and optimized during model training through backpropagation and gradient descent. Fig. 9 illustrates the value variations of α and β throughout the training process. As shown, parameter α is initialized at 0, with its value progressively increasing during training before stabilizing at 0.82. This stabilization trend aligns with the mAP convergence pattern. This empirically validates that the automatic adjustment of α enables the network to gradually shift its focus from horizontal positional features to vertical positional features during training, thereby improving model accuracy. Parameter β is also initialized at 0, but exhibits negative values in early training stages, indicating the model's initial disregard for spatial position information. Subsequently, β values increase steadily during training, ultimately stabilizing at 0.70.

Table 1 presents the ablation study for parameter α. The results indicate that with a fixed α value, the model learns horizontal and vertical positional features with predetermined attention weights, which leads to suboptimal detection and segmentation performance. In comparison, when α is automatically adjusted during training, the model achieves flexible learning of both horizontal and vertical features, thereby obtaining optimal detection and segmentation accuracy. This further confirms that the automatic adjustment of α effectively enhances model performance.

4.5. Ablation study

Our proposed DKGNet builds upon the Mask R-CNN (He et al., 2017) framework by incorporating ESKN and EGKM, which fuse semantic knowledge and embed geometric knowledge, respectively, enabling more accurate detection and segmentation of vehicle components. To validate the effectiveness of these modules, we conduct comprehensive ablation studies on the VMP dataset.

Table 2 shows that after ESKN is added to the baseline model, both the detection and segmentation mAP of the model improved by 15.2%. After the EGKM is added to the baseline model, the detection and segmentation mAP of the model are improved by 14.2% and 14.1%, respectively. When two modules are added to the model simultaneously,

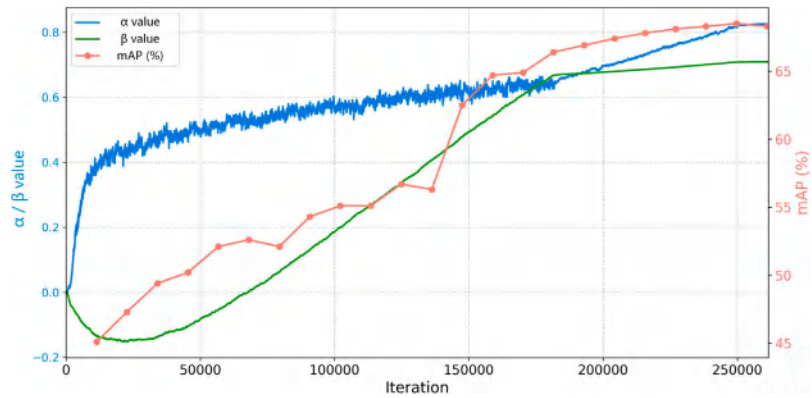


Fig. 9. Parameter variations during training on the VMP dataset.

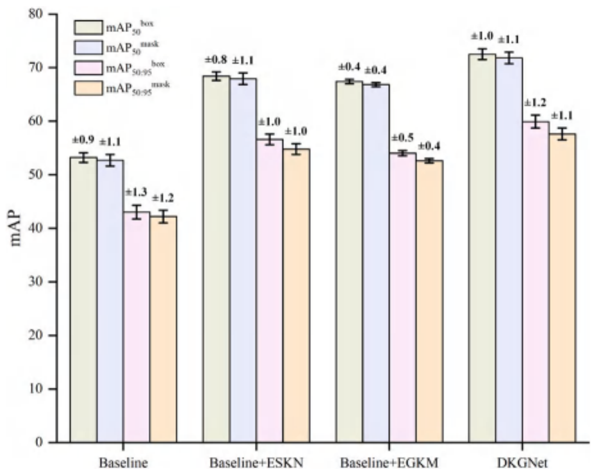


Fig. 10. Confidence intervals of ablation study results for DKGNet on the VMP dataset.

Table 3
Ablation study of ESKN on the VMP dataset.

Method	ESKN		mAP_{50}^{box}	mAP_{50}^{mask}	$mAP_{50:95}^{box}$	$mAP_{50:95}^{mask}$
	SRAM	KSFM				
Mask R-CNN			53.2	52.7	43.0	42.2
	✓		64.8	64.4	51.7	51.1
		✓	58.0	57.6	47.7	46.3
	✓	✓	**68.4**	**67.9**	**56.6**	**54.8**

Table 4
Ablation study of EGKM on the VMP dataset.

Method	EGKM				mAP_{50}^{box}	mAP_{50}^{mask}	$mAP_{50:95}^{box}$	$mAP_{50:95}^{mask}$
	Op-1	Op-2	Op-3	Op-4				
Mask R-CNN					53.2	52.7	43.0	42.2
	✓				56.2	55.6	45.2	44.3
		✓			56.7	56.2	45.8	45.0
			✓		60.8	60.3	48.9	47.6
				✓	53.3	52.8	43.1	42.3
	✓	✓	✓	✓	**67.4**	**66.8**	**54.0**	**52.6**

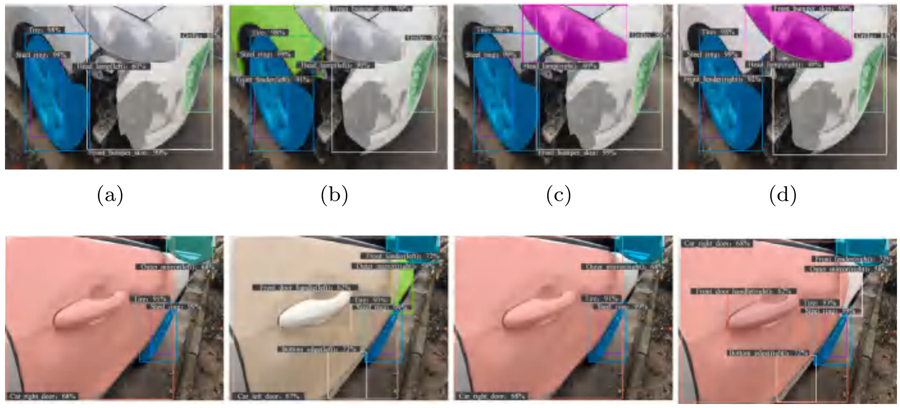
the detection and segmentation mAP of the model increases by 19.3% and 19.1%, respectively. By adding the ESKN and EGKM to the baseline model, we prove that the two modules can enhance the model's vehicle component detection and segmentation abilities to a certain extent, whether used alone or combined. Table 2 presents experimental results calculated as mean values from multiple repeated validations (n = 4). Correspondingly, Fig. 10 visually demonstrates the confidence intervals of the results through error bars, confirming the statistical significance of the data.

Fig. 11 presents the instance segmentation results of the same image processed by Baseline (Fig. 11(a) and Fig. 11(e)), Baseline+ESKN (Figs. 11(b) and 11(f)), Baseline+EGKM (Figs. 11(c) and 11(g)), and DKGNet (Baseline+ESKN+EGKM) (Figs. 11(d) and 11(h)). It can be observed that the model with only ESKN significantly reduces missed detections of components but still suffers from frequent misclassifications of left–right symmetric components. For example, in Fig. 11(b), compared to the baseline, it more comprehensively identifies the Front_fender class but incorrectly classifies it as Front_fender(left). On the other hand, the model with only EGKM accurately distinguishes left–right symmetric components but exhibits a higher rate of missed detections due to its limited learned component classes. For instance, in Fig. 11(g), while it correctly revises Outer_mirror(left) to Outer_mirror(right), it still misses components such as Front_door_handle, Bottom_edge, and Front_fender. By integrating EGKM with ESKN, DKGNet effectively mitigates both misclassifications and missed detections. As shown in Fig. 11(d) and Fig. 11(h), DKGNet achieves more accurate instance segmentation of vehicle components.

In Table 3, we detail the functions of SRAM and KSFM within the ESKN. In Table 4, we provide a detailed overview of the functions of the four operation modules in the EGKM. Table 4 demonstrates that Operation 1 to Operation 4 each consistently enable the model to reliably surpass baseline performance, while the integrated EGKM module combining all four operations achieves significantly superior results compared to any individual operation. As evidenced in Tables 3 and 4, the improvements in each part of both ESKN and EGKM modules collectively enhance DKGNet's performance in vehicle component instance segmentation.

Table 5 provides quantitative experimental validation for the priority selection in EGKM. First, while keeping the auxiliary landmark priorities unchanged and varying the primary landmark priorities, it was found that when the primary landmark components followed the priority order: tire >outer mirror >fender, EGKM achieved its best guidance effect, reaching a detection mAP of 67.4%. Under this priority setting, the model's average inference time per image was 82.2 ms, only 2.8 ms slower than the baseline model. Next, while maintaining the primary landmark priorities and adjusting the auxiliary landmark priorities, the best detection performance and shortest inference time were achieved when the auxiliary landmark components followed the priority order: tail light >door >door handle. Through controlled variable experiments, this study quantitatively validates the rationality of the priority ranking for both primary and auxiliary landmark components in terms of performance metrics.

Fig. 12 displays a comparison of instance segmentation results between the baseline model and DKGNet, evaluated on identical test images from the VMP dataset. We first selected typical vehicle images



(a) (b) (c) (d)

(e) (f) (g) (h)

Fig. 11. Visualization of DKGNet ablation process.

Table 5
Validation of landmark component priority ranking in EGKM on the VMP dataset.

Method	Primary landmark priority	Auxiliary landmark priority	mAP_{50}^{box}	mAP_{50}^{mask}	$mAP_{50:95}^{box}$	$mAP_{50:95}^{mask}$	Time/ Image (ms)
Baseline	None	None	53.2	52.7	43.0	42.2	79.4
Baseline+EGKM	tire >fender >outer mirror	tail light >door >door handle	67.1	66.5	53.8	52.4	82.5
	outer mirror >tire >fender		67.2	66.6	53.9	52.5	82.3
	outer mirror >fender >tire		66.8	66.2	53.4	52.0	85.5
	fender >tire>outer mirror		66.7	66.1	53.3	51.9	89.2
	fender >outer mirror >tire		66.8	66.2	53.4	52.0	92.4
	tire >outer mirror >fender		**67.4**	**66.8**	**54.0**	**52.6**	82.2
	tire >outer mirror >fender	tail light >door handle >door	67.2	66.6	53.9	52.5	86.0
		door >tail light >door handle	67.2	66.6	53.9	52.5	83.2
		door >door handle >tail light	67.2	66.6	53.9	52.5	83.5
		door handle >tail light >door	67.2	66.6	53.9	52.5	90.9
		door handle >door >tail light	67.1	66.5	53.8	52.4	82.4
		tail light >door >door handle	**67.4**	**66.8**	**54.0**	**52.6**	82.2

NOTE: Time/Image denotes the time required for the model to predict a single image, where the mean value is calculated from multiple predictions to ensure the reliability of experimental results.

from four views, including front (Fig. 12(a), Fig. 12(e)), rear (Fig. 12(b), Fig. 12(f)), left (Fig. 12(c), Fig. 12(g)), and right (Fig. 12(d), Fig. 12(h)), to demonstrate the prediction results, proving that the proposed method can accurately identify vehicle components across all views. Next, we selected vehicle images with severe damage conditions, including heavy deformation (Fig. 12(i), Fig. 12(m)), missing and scratched (Fig. 12(j), Fig. 12(n)), deformed and scratched (Fig. 12(k), Fig. 12(o)), and misaligned and deformed (Fig. 12(l), Fig. 12(p)), to demonstrate the prediction results, proving that the proposed method can also accurately identify damaged vehicle components.

From Fig. 12(a)- Fig. 12(d), it is evident that the baseline model exhibits a large number of false detections in left and right component classes across vehicle images from different views. For example, in Fig. 12(a), Head_lamp(right) is incorrectly detected as Head_lamp(left), and Outer_mirror(left) is misclassified as Outer_mirror(right). Additionally, some classes are missed, such as Rear_bumper_skin not being detected in Fig. 12(d). The instance segmentation results of DKGNet demonstrate that using DKGNet can effectively reduce the number of missing and false detections, particularly eliminating the problem of false detections in similar left and right component classes. This improvement benefits from the high occurrence frequency of the landmark components selected in the proposed EGKM module, as they can be identified in the vast majority of vehicle images. For vehicle images from different views, EGKM leverages the relative positions between landmark components to determine vehicle orientation and subsequently classify left–right components.

From Fig. 12(i) to Fig. 12(l), it can be observed that a large number of classes in the segmentation results of the baseline model are missing. For instance, in Fig. 12(i), the baseline model only correctly detects three components: License_plate, Front_bumper_skin, and Grille, with the segmentation of Grille being highly inaccurate. Additionally, in the damaged vehicle images, the baseline model demonstrates numerous false detections across various component classes. For example, in Fig. 12(j), the damage area beneath the vehicle is identified as Rear_bumper_skin, and the upper glass is misclassified as Liftgate_shell. The instance segmentation results of DKGNet demonstrate that it maintains a strong capability for accurate component instance segmentation even in severely damaged vehicle images. This is achieved through multi-scale semantic fusion and geometric spatial knowledge guidance. These results further validating the feasibility of DKGNet's application in intelligent vehicle damage assessment systems.

4.6. Visualization

To validate the effectiveness of ESKN in fusing multi-scale semantic knowledge and facilitating the capture of critical information, we conducted comparative experiments on the VMP dataset by feeding identical images into both the baseline model and the ESKN-enhanced model, with class activation maps visualized using Grad-CAM (Selvaraju et al., 2017). Selected results are shown in Fig. 13. Figs. 13(a)–13(e) present the original input images. Figs. 13(f)–13(j) present the results from the baseline model, while Figs. 13(k)–13(o) show those from the model enhanced with ESKN. The baseline results demonstrate that without ESKN, the model struggles to extract key features from densely packed objects in complex backgrounds. This is evidenced by the class activation maps, where the

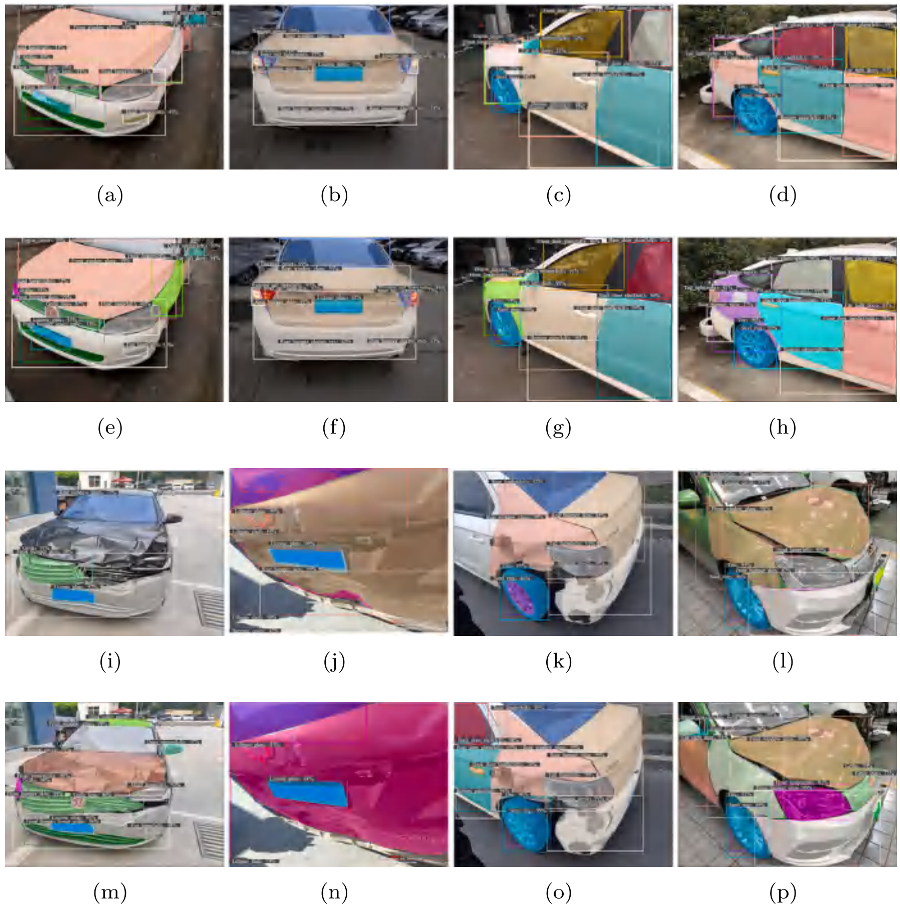


Fig. 12. Comparison between baseline and DKGNet on the VMP dataset: (a)–(d) depict the instance segmentation results of vehicle images under four views based on the baseline model; (e)–(h) depict the instance segmentation results of vehicle images under four views based on DKGNet; (i)–(l) present the instance segmentation results of vehicle images with different damage conditions based on the baseline model; (m)–(p) present the instance segmentation results of vehicle images with different damage conditions based on DKGNet.

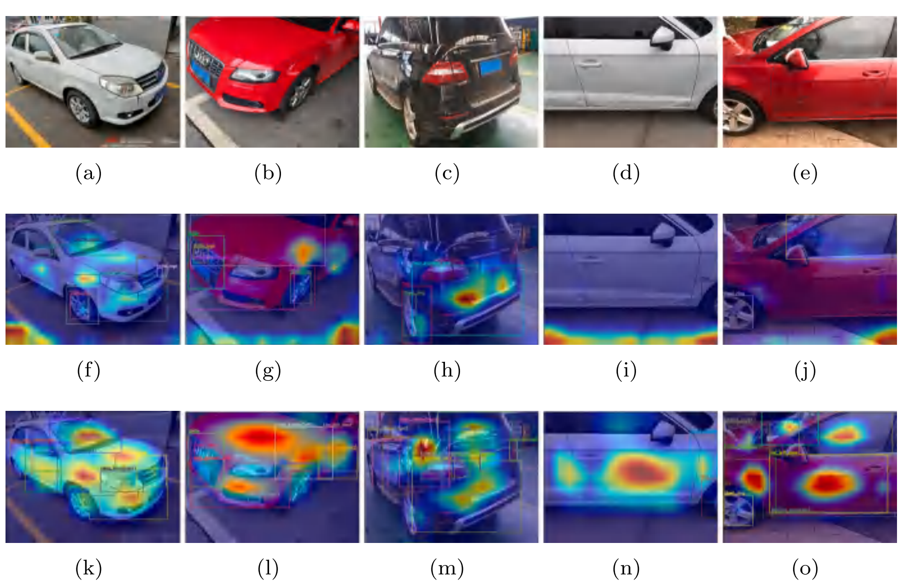


Fig. 13. The visualization of ESKN on the VMP dataset.

baseline model fails to adequately focus on object regions, with its attention significantly deviating from foreground objects. In contrast, Figs. 13(k)–13(o) reveal that the ESKN-enhanced model effectively filters and captures critical object information, precisely attending to both the components themselves and their discriminative inter-class regions. Consequently, it achieves more accurate vehicle component instance segmentation while reducing both missed detections and false positives.

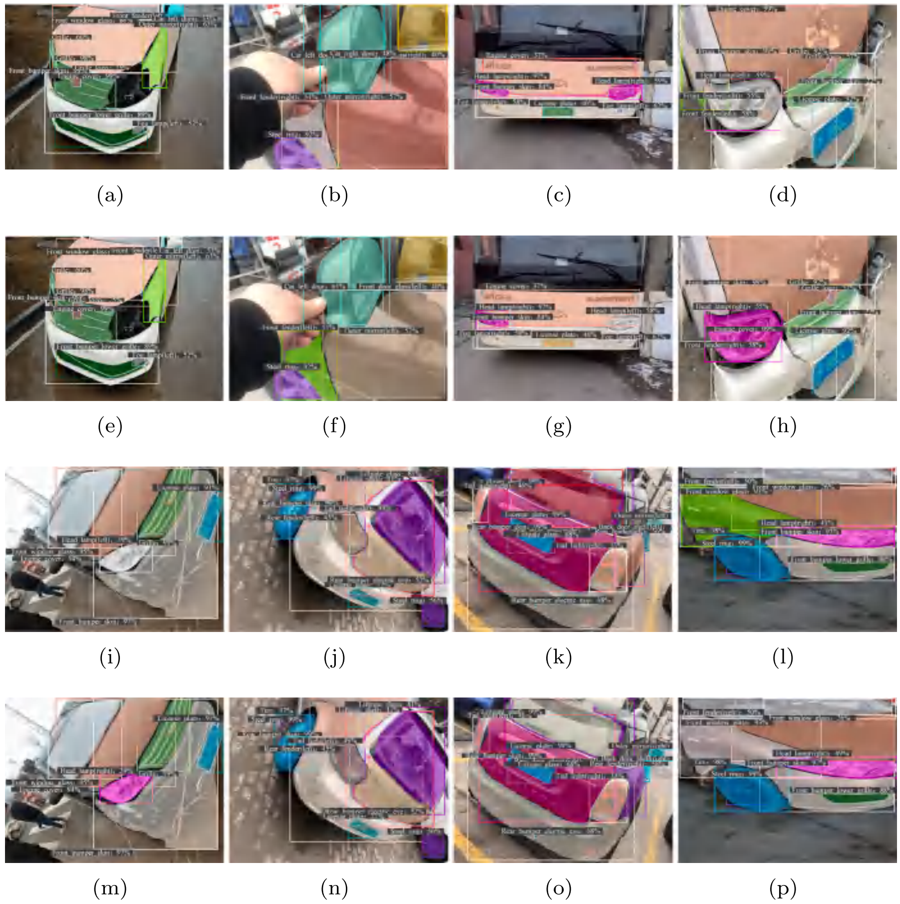


Fig. 14. Robustness experiments of EGKM on the VMP dataset: (a)–(h) show cases when landmark components in EGKM are occluded, missing, or from unconventional vehicles, where (a)–(d) display the detection and segmentation results of the baseline model, and (e)–(h) demonstrate the results after incorporating the EGKM module; (i)–(p) illustrate scenarios with in-plane rotation, camera tilt, or lens distortion, where (i)–(l) present the baseline model's results, and (m)–(p) exhibit the improved results with EGKM.

4.7. Robustness verification

Fig. 14(a)–Fig. 14(h) demonstrates the robustness and interference resistance of the EGKM module when landmark components are occluded, missing, or from unconventional vehicles. In Fig. 14(a) and Fig. 14(e), where components like license plate and tail light are missing, EGKM successfully corrects the mirror classification to Mirror_cover(left) using the grille as a primary landmark component. Figs. 14(b) and 14(f) present cases with front fender occlusion. The comparison between baseline and EGKM results shows that EGKM accurately corrects the classification to Front_fender(left) despite occlusion. Fig. 14(c), Fig. 14(g), Fig. 14(d), and Fig. 14(h) display unconventional vehicle images. As demonstrated, EGKM maintains precise left–right classification even for unconventional vehicles.

The EGKM remains effective even when some landmark components are occluded or missing, due to the diverse categories and comprehensive spatial distribution of selected landmarks. The EGKM can still determine left–right components based on visible landmarks through priority adjustment, even when some landmarks are occluded or missing in the image. The EGKM demonstrates applicability across diverse vehicle types due to the strong representativeness of its selected landmark components, which exhibit minimal variations across different types of vehicles. For vehicle images of various types, including unconventional cases, the EGKM reliably discriminates between left and right components based on the relationships among these landmark components.

Fig. 14(i)–Fig. 14(p) demonstrates the robustness and interference resistance of EGKM when handling in-plane rotation, camera tilt, or lens distortion in vehicle images. Fig. 14(i), Fig. 14(m), Fig. 14(j), and Fig. 14(n) present cases with in-plane vehicle rotation. Comparative results between the baseline and EGKM models show that EGKM successfully corrects left–right component classifications despite image rotation. For instance, in Fig. 14(m), EGKM determines the vehicle's orientation as front orientation and consequently corrects Head_lamp(left) to Head_lamp(right). Fig. 14(k) and Fig. 14(o) illustrate cases with camera tilt during image capture. Visual comparison confirms that camera tilt does not affect EGKM's performance. Figs. 14(l) and 14(p) demonstrate scenarios with lens distortion. The results similarly confirm that lens distortion does not compromise EGKM's functionality.

The effectiveness of the EGKM module under various image perturbations (including in-plane rotation, camera tilt, and lens distortion) stems from the spatial invariance of its embedded geometric knowledge. Specifically, the module relies on stable spatial relationships between components for reasoning. These relative geometric relationships remain consistent even with minor vehicle rotations or deformations in the image, enabling the EGKM to maintain accurate left–right component discrimination.

4.8. Comparison with state-of-the-art methods

4.8.1. Comparison with state-of-the-art attention mechanisms

To verify the superiority of our proposed SRAM module in vehicle component instance segmentation tasks, we conducted comprehensive experiments on the VMP dataset by replacing the attention module

Table 6
Comparison with state-of-the-art attention mechanisms on the VMP dataset.

Method	Attention mechanism					mAP_{50}^{box}	mAP_{50}^{mask}	$mAP_{50:95}^{box}$	$mAP_{50:95}^{mask}$
	ECANet	EPSANet	SegNeXt	LSKNet	SRAM				
Mask R-CNN						53.2	52.7	43.0	42.2
	✓					60.8	60.4	48.5	47.2
		✓				59.8	59.5	48.3	47.1
			✓			62.1	61.8	49.8	48.6
				✓		60.1	59.8	48.3	47.2
					✓	**64.8**	**64.4**	**51.7**	**51.1**

NOTE: The addition positions of the aforementioned attention modules are all the same as the position of SRAM in the proposed method.

Table 7
Comparison with state-of-the-art instance segmentation methods on the VMP dataset.

Method	mAP_{50}^{box}	mAP_{50}^{mask}	$mAP_{50:95}^{box}$	$mAP_{50:95}^{mask}$
Mask R-CNN (He et al., 2017)	53.2	52.7	43.0	42.2
Cascade Mask R-CNN (Cai and Vasconcelos, 2019)	53.7	52.9	44.6	42.5
MS R-CNN (Huang et al., 2019)	54.1	53.2	42.5	42.6
HTC (Chen et al., 2019)	55.3	55.0	46.9	45.0
PointRend (Kirillov et al., 2020)	52.0	51.9	42.4	42.3
SOLOv2 (Wang et al., 2020b)	–	52.4	–	42.3
DCT Mask (Shen et al., 2021)	54.0	53.5	44.6	44.0
Mask Transfiner (Ke et al., 2022)	53.9	53.7	44.6	44.0
FastInst (He et al., 2023)	–	61.1	–	49.9
MTFDN (Zhai et al., 2024b)	59.1	58.9	49.0	47.0
YOLOv12n (Tian et al., 2025)	57.2	56.1	48.7	43.9
DKGNet YOLOv12n	**66.3**	**65.1**	**56.5**	**50.8**
DKGNet Mask R-CNN	**72.5**	**71.8**	**59.9**	**57.6**
DKGNet MTFDN	**78.6**	**77.5**	**64.6**	**60.8**

NOTE: DKGNet YOLOv12n, DKGNet Mask R-CNN, and DKGNet MTFDN refer to the models constructed by incorporating the proposed method into the baseline models of YOLOv12n, Mask R-CNN, and MTFDN, respectively.

in the DKGNet with several state-of-the-art attention mechanisms, including ECANet (Wang et al., 2020a), EPSANet (Zhang et al., 2022), SegNeXt (Guo et al., 2022), and LSKNet (Li et al., 2023). The comparative results with SRAM are presented in Table 6. The experimental results demonstrate that, for vehicle component instance segmentation tasks, SRAM outperforms other attention mechanisms in assisting the model to filter key information among multiple vehicle components, thereby improving both detection and segmentation accuracy.

4.8.2. Comparison with state-of-the-art instance segmentation methods

Our proposed knowledge-guided vehicle component instance segmentation network, DKGNet, integrates the ESKN module (comprising SRAM and KSFM) to effectively fuse multi-scale semantic knowledge from images. Concurrently, DKGNet incorporates geometric knowledge of vehicle components through the EGKM module, which enhances the learning of challenging spatial relationships by analyzing correlations among three types of landmark components. To demonstrate the superiority of DKGNet and evaluate its practical applicability, we conducted comprehensive comparisons with multiple conventional and state-of-the-art instance segmentation methods on the VMP dataset. Meanwhile, to verify the effectiveness and scalability of the dual-knowledge guidance method, we further extended this method to different baseline models for comparative experiments. The experimental results for both detection and segmentation performance are presented in Table 7.

Table 7 presents the comprehensive detection and segmentation performance of all compared methods. The experimental results demonstrate that DKGNet exhibits significant advantages in detecting and segmenting multi-object vehicle components, and the dual-knowledge guidance method can be effectively extended to other instance segmentation algorithms, thereby achieving precise detection and segmentation of vehicle components. Notably, all DKGNet models in this paper employ Mask R-CNN as the baseline. This selection is based on Mask R-CNN's status as a classic algorithm in instance segmentation, whose performance has been extensively validated and established as a universal benchmark for subsequent research. Adopting Mask R-CNN as the baseline model ensures both recognition and reproducibility in experiments.

Table 7 presents experimental results calculated as mean values from multiple repeated validations (n = 4). Correspondingly, Fig. 15 visually demonstrates the confidence intervals of the results through error bars, confirming the statistical significance of the data.

Table 8 compares the detection and segmentation mAP values between the proposed method and baseline models for left–right symmetric vehicle components. The experimental results demonstrate that our method effectively enhances the model's discrimination capability for left–right symmetric components.

Fig. 16 presents a bar chart comparing the segmentation mAP scores of different instance segmentation methods across 59 vehicle component classes on the VMP dataset. The result clearly demonstrates DKGNet's superior segmentation performance for all vehicle component classes.

4.8.3. Generalization verification

To further validate the generalization capability of DKGNet, we compared its detection and segmentation performance with other state-of-the-art instance segmentation methods on the DSMLR car part dataset. Table 9 presents the performance of different instance segmentation methods on the DSMLR car part dataset, showing experimental results calculated as mean values from multiple repeated validations (n=4). Correspondingly, Fig. 17 visually demonstrates the confidence intervals of the results through error bars, confirming the statistical significance of the data. The comparative data in Table 9 show that when faced with state-of-the-art instance segmentation methods of different architectures (CNN-based and Transformer-based), DKGNet achieves the best performance across all accuracy metrics, significantly outperforming the other methods. Specifically, in the bounding box detection task, DKGNet achieves an mAP of 75.0% for 18 classes of vehicle components, representing a 13.3% improvement over the conventional Mask R-CNN. This superior performance primarily benefits from the proposed dual-knowledge collaboration mechanism, where the fusion of semantic knowledge and the embedding of geometric knowledge effectively

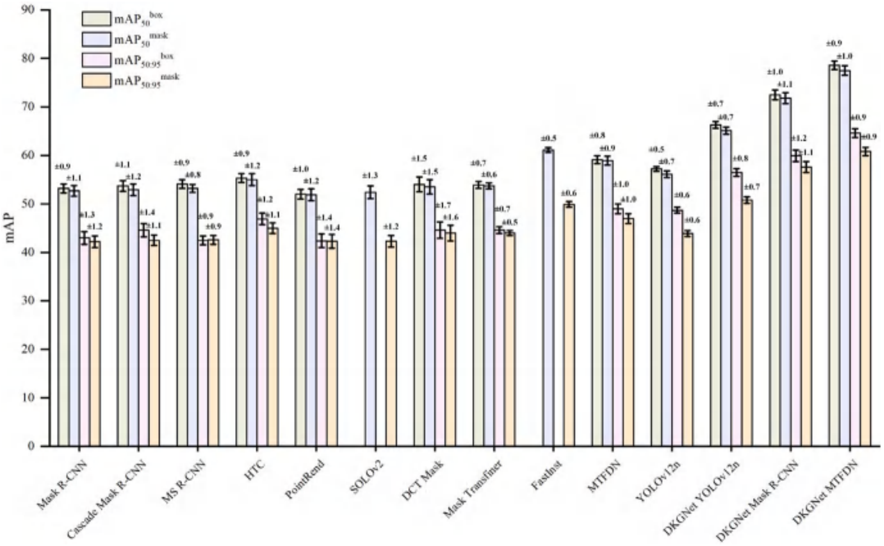


Fig. 15. Confidence intervals of comparison study results on the VMP dataset.

Table 8
Comparison of left–right symmetric components on the VMP dataset.

Labels	YOLOv12n				Mask R-CNN				MTFDN			
	Baseline		DKGNet		Baseline		DKGNet		Baseline		DKGNet	
	mAP_{50}^{box}	mAP_{50}^{mask}	mAP_{50}^{box}	mAP_{50}^{mask}	mAP_{50}^{box}	mAP_{50}^{mask}	mAP_{50}^{box}	mAP_{50}^{mask}	mAP_{50}^{box}	mAP_{50}^{mask}	mAP_{50}^{box}	mAP_{50}^{mask}
2	76.0	51.1	**87.6**	**61.2**	64.3	64.2	**90.6**	**90.7**	63.8	63.9	**89.3**	**89.3**
3	68.3	48.8	**84.4**	**62.2**	59.7	60.2	**89.0**	**89.3**	52.9	53.2	**87.4**	**87.6**
4	27.6	26.3	**33.2**	**31.7**	31.0	30.8	**43.4**	**43.5**	40.3	40.3	**55.9**	**55.9**
5	25.8	25.4	**27.9**	**27.5**	21.7	21.6	**33.9**	**34.1**	24.0	23.8	**45.7**	**45.0**
6	59.1	58.2	**71.6**	**70.8**	51.1	46.7	**73.5**	**67.3**	51.2	47.3	**80.6**	**73.4**
7	44.8	44.0	**63.7**	**63.5**	38.4	36.3	**66.5**	**62.3**	40.9	38.5	**79.1**	**71.3**
12	**33.1**	**32.8**	22.8	22.6	22.1	21.4	**35.5**	**35.0**	40.2	39.5	**50.2**	**42.5**
13	**25.2**	**24.9**	23.0	23.2	22.6	22.2	**35.1**	**34.3**	29.2	29.1	**40.2**	**31.2**
15	41.2	41.2	**58.4**	**58.4**	35.2	35.7	**60.2**	**59.8**	48.9	48.9	**74.5**	**74.5**
16	28.7	28.6	**50.9**	**50.8**	28.5	28.5	**55.8**	**55.7**	39.0	39.3	**70.3**	**70.7**
17	71.5	71.5	**84.5**	**84.5**	64.2	64.3	**82.6**	**82.6**	61.2	61.2	**91.3**	**90.3**
18	58.3	58.2	**78.7**	**78.7**	56.4	56.4	**78.1**	**78.1**	51.1	51.1	**88.2**	**88.3**
19	32.6	31.4	**45.9**	**44.4**	32.5	29.7	**46.6**	**43.3**	48.6	46.6	**72.3**	**66.0**
20	23.2	22.6	**40.0**	**38.6**	21.7	20.9	**41.1**	**38.7**	44.5	44.5	**70.2**	**70.2**
21	44.3	43.8	**59.0**	**58.6**	41.9	41.9	**65.9**	**65.9**	53.1	52.7	**81.2**	**80.5**
22	34.5	34.0	**56.0**	**55.4**	32.0	32.0	**57.9**	**58.0**	42.3	42.2	**73.9**	**73.9**
23	68.4	68.5	**83.1**	**83.2**	53.6	53.8	**83.5**	**83.6**	60.3	60.7	**88.5**	**88.2**
24	65.0	65.6	**81.8**	**82.7**	51.0	51.4	**84.4**	**85.6**	54.4	55.3	**85.5**	**87.6**
25	45.8	45.5	**49.7**	**49.3**	38.0	35.6	**60.3**	**56.0**	53.0	53.0	**70.7**	**68.6**
26	29.1	29.2	**50.7**	**51.0**	26.0	25.3	**57.1**	**57.0**	39.2	39.2	**71.7**	**66.8**
30	**49.2**	**49.1**	49.1	49.0	55.4	55.4	**91.1**	**91.2**	60.1	59.6	**60.1**	**59.6**
31	52.8	52.8	**52.9**	**52.9**	54.6	54.4	**91.4**	**91.2**	59.1	58.9	**59.1**	**58.9**
34	69.6	69.5	**90.8**	**90.6**	52.2	52.3	**95.6**	**95.6**	58.5	58.5	**96.2**	**96.2**
35	67.0	67.0	**89.2**	**89.1**	50.4	50.5	**95.2**	**95.2**	47.9	47.9	**94.9**	**94.9**
37	49.4	49.2	**63.9**	**63.6**	40.4	40.1	**61.9**	**61.5**	46.8	46.5	**73.3**	**73.3**
38	33.3	33.3	**51.7**	**51.7**	26.2	26.2	**57.6**	**57.7**	33.0	33.2	**69.3**	**70.3**
39	73.4	73.4	**85.6**	**85.7**	66.0	67.0	**84.1**	**84.1**	61.1	61.1	**90.7**	**89.6**
40	64.2	64.2	**82.0**	**82.0**	58.2	58.2	**82.2**	**82.2**	51.4	51.4	**90.2**	**90.0**
41	40.8	39.6	**47.5**	**46.5**	32.8	30.2	**52.3**	**50.3**	33.0	32.9	**52.5**	**50.9**
42	16.7	16.3	**32.5**	**32.0**	16.6	15.1	**40.0**	**35.8**	25.7	24.5	**52.1**	**47.6**
43	46.2	45.7	**58.5**	**57.9**	56.1	55.9	**64.6**	**64.0**	57.1	57.1	**78.1**	**78.0**
44	29.7	29.0	**51.8**	**50.8**	49.8	49.7	**56.5**	**56.5**	50.6	50.5	**75.2**	**74.3**
45	39.5	39.7	**50.0**	**50.4**	39.1	39.4	**73.0**	**73.9**	45.7	46.4	**77.1**	**77.5**
46	40.2	40.6	**48.6**	**49.0**	39.0	38.3	**77.0**	**77.5**	34.8	35.4	**71.8**	**72.9**
47	74.4	74.6	**90.9**	**91.0**	51.6	51.5	**89.2**	**89.2**	59.7	59.8	**94.7**	**94.8**
48	69.8	69.9	**90.0**	**90.2**	48.4	48.4	**89.9**	**90.0**	53.2	53.4	**93.6**	**93.6**
49	40.9	40.9	**52.4**	**52.4**	32.2	30.9	**55.0**	**52.4**	48.8	47.8	**69.2**	**67.4**
50	30.1	30.0	**49.5**	**49.4**	20.4	18.0	**54.2**	**46.1**	27.3	24.0	**69.6**	**63.7**
51	47.3	47.3	**59.7**	**59.7**	52.9	52.9	**87.5**	**87.5**	57.3	57.3	**84.8**	**84.8**
52	52.5	52.5	**61.5**	**61.5**	53.1	53.1	**90.8**	**90.8**	59.5	59.5	**86.2**	**86.2**
53	48.8	48.8	**63.7**	**63.7**	45.5	45.5	**82.2**	**82.3**	54.2	54.1	**82.6**	**82.0**
54	50.7	50.7	**63.5**	**63.4**	47.4	47.1	**82.1**	**82.0**	48.7	48.7	**75.2**	**75.2**

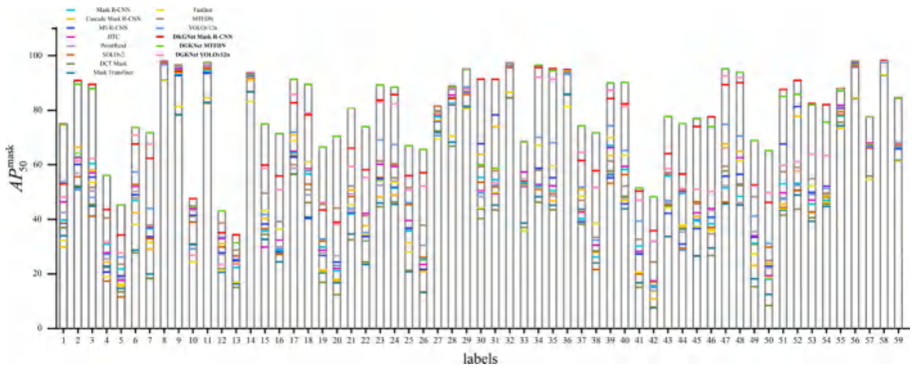


Fig. 16. Per-class segmentation performance of instance segmentation methods on the VMP dataset.

Table 9
Comparison with state-of-the-art instance segmentation methods on the DSMLR car part dataset.

Method	mAP_{50}^{box}	mAP_{50}^{mask}	$mAP_{50:95}^{box}$	$mAP_{50:95}^{mask}$
Mask R-CNN (He et al., 2017)	61.7	62.4	38.0	40.4
Cascade Mask R-CNN (Cai and Vasconcelos, 2019)	63.8	63.4	43.7	42.4
MS R-CNN (Huang et al., 2019)	64.5	64.5	39.8	43.2
HTC (Chen et al., 2019)	60.6	60.4	38.2	38.8
PointRend (Kirillov et al., 2020)	61.5	61.4	36.8	43.3
SOLOv2 (Wang et al., 2020b)	–	60.4	–	36.4
DCT Mask (Shen et al., 2021)	61.2	62.7	39.9	44.4
Mask Transfiner (Ke et al., 2022)	61.5	62.4	39.9	43.2
Mask2Former (Cheng et al., 2022)	58.4	65.2	37.2	41.1
FastInst (He et al., 2023)	–	66.0	–	48.3
DKGNet Mask R-CNN	**75.0**	**74.2**	**46.8**	**49.7**

enhance the model's discriminative capability for multi-object vehicle components. The experimental results fully demonstrate that DKGNet, through the dual application of semantic and geometric knowledge, can effectively address the inherent challenges in vehicle component recognition, including large intra-class feature variations, and small inter-class feature differences, while exhibiting strong generalization capability.

DKGNet can generalize to datasets with different annotation standards for two main reasons. First, through multi-scale semantic fusion, it captures salient semantic information across diverse datasets. Second, the geometric prior knowledge employed by DKGNet is based on the relatively stable spatial relationships between vehicle components, and this spatial relativity remains consistent across datasets. These two factors enable DKGNet to maintain stable performance even when faced with datasets featuring varying annotation standards.

Table 10 presents a comparison of computational requirements among different instance segmentation methods on the DSMLR car part dataset. It can be observed that compared to some baseline models, the proposed method exhibits moderate increases in parameter count, FLOPs, and inference latency. However, this represents a reasonable trade-off given its substantial improvements in both detection and segmentation accuracy. These modest computational overheads are considered acceptable for the task of intelligent vehicle damage assessment.

4.9. Limitations

The proposed method performs well for vehicle images at medium and long distances, but shows limited effectiveness for extremely localized vehicle images. As shown in Fig. 18, when the vehicle content in the image is too localized or captured from extreme angles, the insufficient number of objects providing structural and positional information prevents the full utilization of both semantic and geometric knowledge, ultimately leading to inaccurate identification of vehicle components. For instance, in Fig. 18(a), Front_fender(left) is misclassified as Front_fender(right), while in Fig. 18(b), Outer_mirror(right) is misclassified as Outer_mirror(left).

5. Conclusion

To achieve more accurate detection and segmentation of vehicle components, we propose Dual-Knowledge Guided vehicle component instance segmentation Network (DKGNet). Specifically, to address the multi-object challenges and the issue of low intra-class similarity but high inter-class similarity in vehicle component instance segmentation, we introduce the enhanced semantic knowledge network, which enables the model to filter and capture detailed foreground information and complex features across different scales through multi-scale semantic knowledge fusion. Additionally, we propose the embedded geometric knowledge module, which integrates model performance with spatial geometric knowledge of vehicle components, facilitating precise classification of mirror-symmetric classes. In comparison with

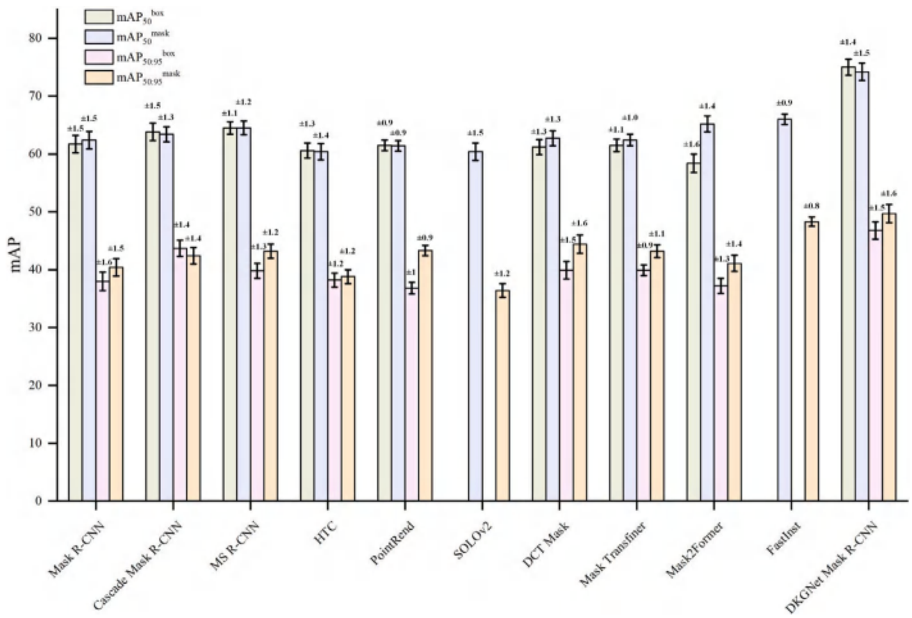


Fig. 17. Confidence intervals of comparison study results on the DSMLR car part dataset.

Table 10
Computational requirements of different instance segmentation methods on the DSMLR car part dataset.

Method	Parameters (M)	FLOPs (G)	Inference Latency (ms)
Mask R-CNN (He et al., 2017)	44.3	202.8	38.7
MS R-CNN (Huang et al., 2019)	60.4	269.1	36.2
PointRend (Kirillov et al., 2020)	59.9	177.3	34.9
SOLOv2 (Wang et al., 2020b)	46.6	219.1	39.4
Mask Transfiner (Ke et al., 2022)	54.6	665.3	55.1
Mask2Former (Cheng et al., 2022)	47.4	224.4	88.1
DKGNet Mask R-CNN	58.7	231.3	85.8

NOTE: All experiments were conducted on a single NVIDIA GeForce GTX 1080Ti GPU.

(a) (b)

Fig. 18. Cases where DKGNet fails to accurately identify vehicle components.

state-of-the-art instance segmentation methods, DKGNet demonstrates significant advantages in the detection and segmentation of vehicle components, particularly in instance segmentation of mirror-symmetric left–right components. DKGNet can provide practical guidance for the design of intelligent vehicle damage assessment systems.

However, our method cannot achieve accurate recognition of vehicle components in highly localized images. In future work, we should investigate the mutual matching of distant-view and close-view images. This alignment of global and local information across views will enable precise identification of vehicle components in localized images. Furthermore, given sufficient training data, the proposed knowledge-guided method can be integrated with foundational image segmentation models to enhance their adaptability to the vehicle domain, ultimately achieving superior performance in vehicle component segmentation.

CRediT authorship contribution statement

Zhenqi Zhang: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Funding acquisition, Conceptualization. **Xunqi Zhou:** Writing – review & editing, Writing – original draft, Visualization, Validation, Formal analysis, Data curation. **Yongjie Zhai:** Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Nianhao Chen:** Investigation, Data curation. **Qianming Wang:** Writing – original draft, Resources, Project administration. **Xinying Wang:** Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

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